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ScienceDirect
Research in Veterinary Science 84 (2008) 159–165
www.elsevier.com/locate/rvsc

A proposed transpositional acupoint system in a mouse and rat model

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Accepted 6 April 2007

Abstract

In acupuncture, the specificity of the point and the reproducibility of the location of the point are prerequisite to the specificity and reproducibility of research involving acupuncture stimulation. The transpositional method, which locates animal acupoints (AAs) on the surface of animal skin corresponding to the anatomic site of a human acupoint, has been generally adopted for research modeling. However, there remains a lack of consensus on the specific location of AA among researchers. The potential problems that the discrepancy in acupoint locating methods causes include the attempt to compare research results. This report is a conceptual study that calls attention to the problems of inconsistency in AA location, and proposes a transpositional 121 AA system in a mouse and rat model. Further discussion, and the establishment of reproducible transpositional AA systems, will prompt further quantitative research and exchange of scientific ideas.
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Keywords: Acupuncture points; Animal acupoints; Mouse; Rat; Transpositional method

1. Introduction

Acupuncture involves insertion of a needle into a specific point on the skin. Therefore, the description of the specificity of the point and the reproducibility of the location of the insertion point are prerequisite to the specificity and reproducibility of the research involving acupuncture stimulation.

Research and clinical application of acupuncture are pursued in both the fields of human medicine and veterinary medicine. Animal studies are an indispensable part of acupuncture research relied on to elucidate the mechanism of acupuncture effects. An increasing number of acupuncture studies are reported using acupoints of laboratory animals such as the mouse and rat.

The transpositional method, which locates animal acupoints (AAs) on the surface of their skin, corresponding to the anatomic site of a human acupoint, has been a technique generally adopted by many researchers; transpositional acupoint systems of the rat and mouse were presented by Koh (1999, 2001). However, there is still no consensus on specific locations of AA among researchers. The resultant discrepancy of acupoint location results in

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doi:10.1016/j.rvsc.2007.04.004

difficulty in interpreting and comparing studies reported in the literature.

Lack of a consensus on the location of AAs among investigators is an important problem that may be restricting scientific advancement. Therefore, in this paper, a revision of the transpositional acupoint system of the mouse and rat is presented for specific, reproducible location of AA, and discussion of the application of this AA locating method is presented.

2. Materials and methods

To develop a transpositional acupoint system for the mouse and rat, we considered the following principles: (1) For the sake of practical application, the proposed transpositional AA system will include only part of the meridian acupoints that have been used frequently in laboratory studies or clinical practice based on previous reports (Koh, 1999, 2001), rather than including all of the 361 human meridian points. Frequently used, main acupoints of 14 main meridians are included. (2) The nomenclature of individual acupoints, of the proposed AA system, follows WHO nomenclature (WHO, 1993). (3) The proposed transpositional AAs will be located at the point that anatomically corresponds to the human acupoint. (4) The anatomical differences between human and animals will be considered. The anatomic difference between the human and the laboratory animals, such as the mouse and the rat,

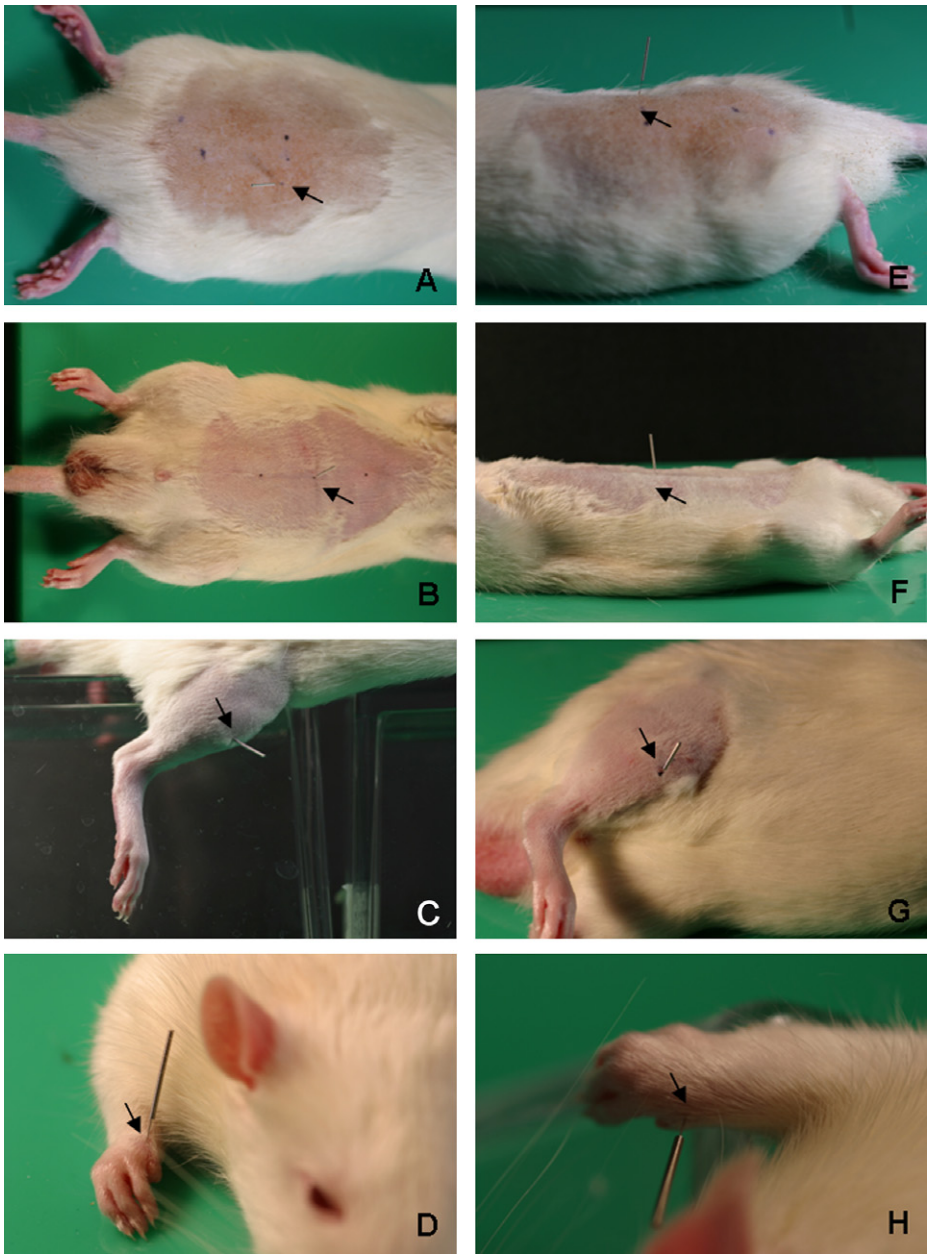

Fig. 1. Acupuncture needle is inserted into the animal acupoints of BL23 (A, E), CV12 (B, F), ST36 (C, G), and LI4 (D, H).

prevents a simple translation from animal anatomy to human anatomy. For example the liver, gall bladder, spleen, and stomach are located below the ribcage within the abdominal cavity under the diaphragm of humans, while in mice and rats some organs are within the ribcage under the diaphragm. In addition, the rat does not have a gall bladder. And the human and the mouse or rat differs in the numbers of thoracic vertebrae. Therefore, BL18, BL19, BL20 and BL21, that is, the points associated with those organs are proposed to be near the abdominal cavity, at the levels of the last four ribs. (5) The proportional measures, of AA location, are adjusted to simpler measures in the transpositional AA system taking into consideration practical application and the original meaning of the acupoints. The proportional method for acupoint location describes a specific surface point on an imaginary line connecting two landmarks using the relative distances from both landmarks. The mouse and the rat are relatively small. The proportions used in the human acupoint system are too complex for comparisons with small animals and are not always applicable to AAs. (6) Finally, the acupoint GV20L was added to the transpositional AA system considering the original description of the acupoint GV20. The GV20 acupoint is at the human vertex and is the uppermost acupoint. The human is erect but laboratory animals are quadruped. Therefore, an additional acupoint GV20L was included in the proposed AA system, the location of which was proposed to be at the uppermost lumbar site between the last lumbar vertebra and the sacrum of the rat or mouse (Koh and Yin, 1999) (Fig. 1).

3. Results

3.1. Proposed acupoint system for the rat and mouse

3.1.1. Lung meridian (LU)

LU5: is located at the depression lateral to the biceps brachii tendon, in the cubital fossa.
LU9: is located at the depression medial to the extensor carpi radialis tendon, on the medial side of wrist.
LU11: is located at the medial side of the tip of the pollex, in the forefoot.

3.1.2. Large intestine meridian (LI)

LI1: is located at the medial side of the tip of the first digit, in the forefoot.
LI4: is located on the first dorsal interossei, medial to the middle of the second metacarpal bone, in the forefoot.
LI11: is located at the depression medial to the extensor carpi radialis, at the lateral end of cubital crease.
LI15: is located at the depression anterior–inferior to the acromial end, on the anterior side of deltoid.
LI20: is located just lateral to the ala nasi.

3.1.3. Stomach meridian (ST)

ST1: is located at the depression inferior to the eyeball on the mid pupil line, superior to the lower margin of the orbital fossa.
ST2: is located at the infraorbital foramen.
ST6: is located on the muscle belly, anterior–superior side of mandibular angle.
ST7: is located at the depression inferior to the zygomatic arch, anterior to the mandibular condyle.
ST25: is located lateral to the umbilicus, at the midpoint from the umbilicus to the nipple line.
ST35: is located at the depression lateral to the patella ligament.
ST36: is located at the proximal one fifth point on the line from ST35 to the anterior side of ankle crease.
ST40: is located at the midpoint from ST35 to the lateral malleolus of tibiofibula.
ST41: is located at the depression between tibialis anterior tendon and extensor digitorum longus tendon, on the anterior side of ankle crease.
ST45: is located at the lateral side of the tip of the second toe, in the hindfoot.

3.1.4. Spleen meridian (SP)

SP1: is located at the medial side of the tip of the hallux, in the hindfoot.
SP2: is located at the depression anterior to the first metatarsophalangeal joint, on the medial side of hindfoot.
SP3: is located at the depression posterior to the first metatarsophalangeal joint, on the medial side of hindfoot.
SP4: is located at the depression anterior to the base of first metatarsal bone, on the medial side of hindfoot.
SP5: is located at the depression medial to the tibialis anterior tendon, inferior to the medial malleolus of tibiofibula.
SP6: is located at the proximal endpoint of the distal one fifth on the imaginary line connecting SP9 and the medial malleolus of tibiofibula.
SP9: is located at the depression inferior to the medial condyle of the tibia, between tibia and gastrocnemius.
SP10: is located at the depression superior to the medial condyle of the femur, on the medial side of thigh.
SP21: is located at the depression below the sixth rib, on the mid axillary line.

3.1.5. Heart meridian (HT)

HT1: is located at the depression in the middle of the axilla.
HT3: is located at the depression anterior to the triceps brachii tendon, between the medial epicondyle of the humerus and the olecranon process.

HT7: is located at the depression on the posterior lateral side of the wrist, between flexor carpi ulnaris tendon and flexor digitorum superficialis tendon.
HT9: is located at the medial side of the tip of the fourth digit, in the forefoot.

3.1.6. Small intestine meridian (SI)

SI1: is located at the lateral side of the tip of the fourth digit, in the forefoot.
SI5: is located at the depression between the carpal bone and the ulna, on the lateral side of the wrist joint.
SI8: is located at the depression between the lateral epicondyle of the humerus and the olecranon process.
SI19: is located at the depression anterior to the tragus.

3.1.7. Bladder meridian (BL)

BL1: is located at the point medial superior to the medial canthus.
BL10: is located at the depression on the posterior border of the lateral mass of the atlas.
BL11: is located at the depression lateral to the lower border of spinous process of the first thoracic vertebra (approximately 0.8 cm in the adult rat and 0.3 cm in the adult mouse, approximately one twentieth of the circumference of the thorax).
BL13: is located at the depression lateral to the lower border of spinous process of the third thoracic vertebra (approximately 0.8 cm in the adult rat and 0.3 cm in the adult mouse, approximately one twentieth of the circumference of the thorax).
BL14: is located at the depression lateral to the lower border of spinous process of the fourth thoracic vertebra (approximately 0.8 cm in the adult rat and 0.3 cm in the adult mouse, approximately one twentieth of the circumference of the thorax).
BL15: is located at the depression lateral to the lower border of spinous process of the fifth thoracic vertebra (approximately 0.8 cm in the adult rat and 0.3 cm in the adult mouse, approximately one twentieth of the circumference of the thorax).
BL16: is located at the depression lateral to the lower border of spinous process of the sixth thoracic vertebra (approximately 0.8 cm in the adult rat and 0.3 cm in the adult mouse, approximately one twentieth of the circumference of the thorax).
BL17: is located at the depression lateral to the lower border of spinous process of the seventh thoracic vertebra (approximately 0.8 cm in the adult rat and 0.3 cm in the adult mouse, approximately one twentieth of the circumference of the thorax).
BL18: is located at the depression lateral to the lower border of spinous process of the tenth thoracic vertebra (approximately 0.8 cm in the adult rat and 0.3 cm in the adult mouse, approximately one twentieth of the circumference of the thorax).
BL19: is located at the depression lateral to the lower border of spinous process of the eleventh thoracic vertebra (approximately 0.8 cm in the adult rat and 0.3 cm in the adult mouse, approximately one twentieth of the circumference of the thorax).
BL20: is located at the depression lateral to the lower border of spinous process of the twelfth thoracic vertebra (approximately 0.8 cm in the adult rat and 0.3 cm in the adult mouse, approximately one twentieth of the circumference of the thorax).
BL21: is located at the depression lateral to the lower border of spinous process of the thirteenth thoracic vertebra (approximately 0.8 cm in the adult rat and 0.3 cm in the adult mouse, approximately one twentieth of the circumference of the thorax).
BL22: is located at the depression lateral to the lower border of spinous process of the first lumbar vertebra (approximately 0.8 cm in the adult rat and 0.3 cm in the adult mouse, approximately one twentieth of the circumference of the thorax).
BL23: is located at the depression lateral to the lower border of spinous process of the second lumbar vertebra (approximately 0.8 cm in the adult rat and 0.3 cm in the adult mouse, approximately one twentieth of the circumference of the thorax).
BL24: is located at the depression lateral to the lower border of spinous process of the third lumbar vertebra (approximately 0.8 cm in the adult rat and 0.3 cm in the adult mouse, approximately one twentieth of the circumference of the thorax).
BL25: is located at the depression lateral to the lower border of spinous process of the fourth lumbar vertebra (approximately 0.8 cm in the adult rat and 0.3 cm in the adult mouse, approximately one twentieth of the circumference of the thorax).
BL26: is located at the depression lateral to the lower border of spinous process of the fifth lumbar vertebra (approximately 0.8 cm in the adult rat and 0.3 cm in the adult mouse, approximately one twentieth of the circumference of the thorax).
BL27: is located at the depression lateral to the lower border of spinous process of the sixth lumbar vertebra (approximately 0.8 cm in the adult rat and 0.3 cm in the adult mouse, approximately one twentieth of the circumference of the thorax).
BL28: is located at the depression lateral to the imaginary point located inferior to BL27 by the length of the line connecting BL26 and BL27 (approximately 0.8 cm in the adult rat and 0.3 cm in the adult mouse, approximately one twentieth of the circumference of the thorax).
BL35: is located at the depression lateral to the base of the tail.
BL39: is located at the depression in the lateral end of popliteal fossa, medial to the biceps femoris tendon.
BL40: is located at the depression in the middle of the popliteal fossa.

BL52: is located at the depression lateral to the lower border of spinous process of the second lumbar vertebra (approximately 0.8 cm in the adult rat and 0.3 cm in the adult mouse, approximately one twentieth of the circumference of the thorax).
BL57: is located at the midpoint between BL40 and BL60.
BL60: is located at the depression posterior to the lateral malleolus of tibiofibula.
BL67: is located at the lateral side of the tip of the fifth toe, in the hindfoot.

3.1.8. Kidney meridian (KI)

KI1: is located at the depression in the middle of the metatarsal area, on the plantar side.
KI3: is located at the depression posterior to the medial malleolus of tibiofibula.

3.1.9. Pericardium meridian (PC)

PC3: is located at the depression medial to the biceps brachii, on the cubital fossa.
PC6: is located at the depression of the distal 5/6 point on the imaginary line connecting PC3 and PC7, posterior to the flexor carpi radialis tendon, anterior to the flexor digitorum superficialis tendon.
PC7: is located at the depression just superior to the carpal bones, posterior to the flexor carpi radialis tendon, on the palmar side of wrist crease.
PC9: is located at the medial side of the tip of the second finger, in the forefoot.

3.1.10. Triple energizer meridian (TE)

TE1: is located at the lateral side of the tip of the third finger, in the forefoot.
TE4: is located at the depression posterior to the extensor digitorum communis tendon, on the middle of the dorsal surface of the wrist crease.
TE5: is located at the radioulnar depression of the distal 5/6 point on the imaginary line connecting wrist and elbow, posterior to the extensor digitorum communis tendon.
TE10: is located at the depression superior to the olecranon process.

3.1.11. Gall bladder meridian (GB)

GB1: is located at the point lateral to the lateral canthus.
GB3: is located at the depression superior to the zygomatic arch, vertically corresponding to ST7.
GB20: is located at the depression medial to the insertion of sternomastoid and cleidomastoid, on the level of GV16.
GB21: is located at the midpoint of the line connecting the seventh cervical vertebra and the acromion.
GB25: is located at the depression posterior to the end of thirteenth rib, on the flank area.
GB30: is located at the depression superior to the greater trochanter of the femur.
GB33: is located at the depression superior to the lateral condyle of the femur, on the lateral side of thigh.
GB34: is located at the depression anterior and inferior to the fibular head.
GB39: is located at the distal 4/5 point on the line connecting the lateral side of the knee and the lateral malleolus of the tibiofibula.
GB40: is located at the depression anterior and inferior to the lateral malleolus of tibiofibula.
GB44: is located at the lateral side of the tip of the fourth toe, in the hindfoot.

3.1.12. Liver meridian (LR)

LR1: is located at the lateral side of the tip of the hallux, in the hindfoot.
LR2: is located at the depression anterior to the midpoint between the first and second metatarsophalangeal joints.
LR3: is located at the proximal end of the depression between the first and second metatarsal bones.
LR5: is located at the distal 2/3 point on the line connecting the medial side of the knee and the medial malleolus of the tibiofibula.
LR8: is located at the depression anterior to the insertion of the semitendinosus and semimembranosus tendons, on the medial side of the knee.
LR13: is located at the depression posterior to the end of twelfth rib, on the flank area.
LR14: is located at the depression below the sixth rib, on the nipple line.

3.1.13. Conception vessel meridian (CV)

CV1: is located at the midpoint depression between anus and pudendum.
CV3: is located at the point of 4/5 down the ventral midline connecting the umbilicus to the pubic tubercle.
CV4: is located at the point of 3/5 down the ventral midline connecting the umbilicus to the pubic tubercle.
CV6: is located at the point of 3/10 down the ventral midline connecting the umbilicus to the pubic tubercle.
CV7: is located at the point of 1/5 down the ventral midline connecting the umbilicus to the pubic tubercle.
CV9: is located at the point of 1/8 up the ventral midline connecting the umbilicus to the sternum.
CV10: is located at the point of 2/8 up the ventral midline connecting the umbilicus to the sternum.
CV11: is located at the point of 3/8 up the ventral midline connecting the umbilicus to the sternum.
CV12: is located at the mid point of the ventral midline connecting the umbilicus to the sternum.

CV13: is located at the point of 5/8 up the ventral midline connecting the umbilicus to the sternum.
CV14: is located at the point of 6/8 up the ventral midline connecting the umbilicus to the sternum.
CV15: is located at the point of 7/8 up the ventral midline connecting the umbilicus to the sternum.
CV17: is located at the level of fourth intercostals space, on the ventral midline.

3.1.14. Governing vessel meridian (GV)

GV1: is located at the depression between the base of the tail and the anus.
GV2: is located at the depression between the sacrum and the first coccygeal vertebra, on the dorsal midline.
GV20L: is located at the depression between the sixth lumbar vertebra and the sacrum, on the dorsal midline.
GV3: is located at the depression between the spinous processes of the fourth and the fifth lumbar vertebrae, on the dorsal midline.
GV4: is located at the depression between the spinous processes of the second and the third lumbar vertebrae, on the dorsal midline.
GV5: is located at the depression between the spinous processes of the first and the second lumbar vertebrae, on the dorsal midline.
GV6: is located at the depression between the spinous processes of the twelfth and the thirteenth thoracic vertebrae, on the dorsal midline.
GV7: is located at the depression between the spinous processes of the eleventh and the twelfth thoracic vertebrae, on the dorsal midline.
GV8: is located at the depression between the spinous processes of the tenth and the eleventh thoracic vertebrae, on the dorsal midline.
GV9: is located at the depression between the spinous processes of the seventh and the eighth thoracic vertebrae, on the dorsal midline.
GV10: is located at the depression between the spinous processes of the sixth and the seventh thoracic vertebrae, on the dorsal midline.
GV11: is located at the depression between the spinous processes of the fifth and the sixth thoracic vertebrae, on the dorsal midline.
GV12: is located at the depression between the spinous processes of the third and the fourth thoracic vertebrae, on the dorsal midline.
GV13: is located at the depression between the spinous processes of the first and the second thoracic vertebrae, on the dorsal midline.
GV14: is located at the depression between the spinous processes of the seventh cervical and the first thoracic vertebrae, on the dorsal midline.
GV16: is located at the depression between the occiput and the atlas, on the dorsal midline.
GV20: is located at the vertex, on the dorsal midline.
GV25: is located at the tip of the nose, at the margin of the haired skin area, on the facial midline.
GV26: is located at the mid point between base of the columellar nasi and the upper lip, on the facial midline.

4. Discussion

4.1. Three methods to determine the specific location of AA

There exist three methods by which AAs are specifically located. First, the AAs of the traditional veterinary acupuncture system may be adopted. Traditional veterinary acupuncture has its own long history, and there are numerous descriptions of the AAs in the literature (Koh and Yin, 1999). In the past, the animals treated with traditional veterinary acupuncture were mainly agricultural animals including horses, cows, pigs and poultry (Schoen, 1994). The laboratory animals such as the mouse and rat were used in biomedical research. There is no description of the AAs of these animals in traditional veterinary acupuncture literature. Therefore, this is the first method presented for application of AA to these small laboratory animals. Second, AAs may also be located by physiologic characteristics thought to be characteristic of acupoints, for example, reduced electrical impedance (Kawakita et al., 1991) or excitable muscle/skin-nerve complexes with high density of nerve endings (Li et al., 2004). However, there remains controversy with regard to the anatomic sites related to acupoints and meridians. It has not yet been established whether the so-called physiologic characteristics of acupoints reflect the acupoints accurately. There is no new system to explain acupoint and meridian that replaces the traditional concept of acupoint and meridian. The third is the transpositional method, which locates the veterinary acupoints by transposing human acupuncture points onto animal anatomy (Panzer, 1993). In reality, this third method is a widely acknowledged and adopted technique used in laboratory animals.

4.2. Discrepancy of the location of the animal acupoint in published reports

There remains no consensus on the location of AAs in laboratory animals among researchers, even when they are determined by the same method of transpositional location. For example, the ST36 point is one of the most frequently used points in acupuncture practice, and also in acupuncture research. However, the ST36 site in the laboratory rat has different meaning for different researchers; in recent reports, its site has been referred to as 5 mm lower and lateral to the anterior tubercle of tibia (Kim et al., 2002, 2003), or near the knee joint of the hind limb, 2 mm lateral to the anterior tubercle of the tibia (Yeom et al., 2003; Lim et al., 2003). Such a discrepancy precludes

the ability of researchers to compare their results, which imposes serious limitations on scientific communication.

4.3. Difference of anatomy between animal and human

The presupposition of the transpositional method of AA location is that acupoint charts may be drawn for all vertebrate animals, based on the human model (Panzer, 1993). The similarity of anatomic structures is the basis of the transposing approach. The difference in anatomy may prevent a simple translation from human anatomy to animal anatomy. While the human has 12 thoracic vertebrae, five lumbar vertebrae, five sacral vertebrae and three to six coccygeal vertebrae, the rat has 13 thoracic, six lumbar, four sacral, and 28 coccygeal vertebrae. While the human has two floating ribs, the rat has three floating ribs. In addition, the position of the scapulae or the iliums in relation to the extremities is different. The radius is in the pronated position in the rat.

Laboratory animals are relatively small. The proportional method for the human acupoint location is accepted as the most reliable method for human acupoint location compared to others (Yin et al., 2005); however, it may be too complex for small laboratory animals, making it not applicable to all the transpositional acupoints. According to the human acupoint location system, for example, the distance between ST35 and ST41 is 16 cun and the ST36 is located at 3/16 point of the imaginary line drawn from ST35 to ST41. The distance from ST35 to ST41 in three Sprague–Dawley rats weighing 298–310 g ranged from 35 to 37 mm. It was not practical to locate ST36 at the 3/16 point from ST35 to ST41. Therefore, simpler measures for the proportional method of acupoint location, the 1/5 point from ST35 to ST41 as ST36, for example, were considered. We proposed simpler measures considering practical application and the original meaning of the acupoints. The WHO scientific study group suggested that a new point must be at least 0.5 cun away from a classical acupuncture point (WHO, 1991). Simpler measures, here proposed, do not preclude the above mentioned principle. The distance between the 3/16 point from ST35 to ST41 and the 1/5 point from ST35 to ST41 is less than 1 mm, which is fairly within the range of 0.5 cun difference.

4.4. Proposed transpositional acupoint systems of the rat and the mouse

We organized and proposed transpositional acupoint systems for the rat and the mouse based upon previous reports (Koh, 1999, 2001). These systems are, of course, not a final consensus. We hope our proposal will provide momentum to draw the attention of investigators to the need for a consensus on acupoint location that will serve as a basis for accurate and precise communication and the further development of the science of acupuncture medicine.

Acknowledgement

This study was supported by the SRC Program of KOSEF (R11-2005-014).

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